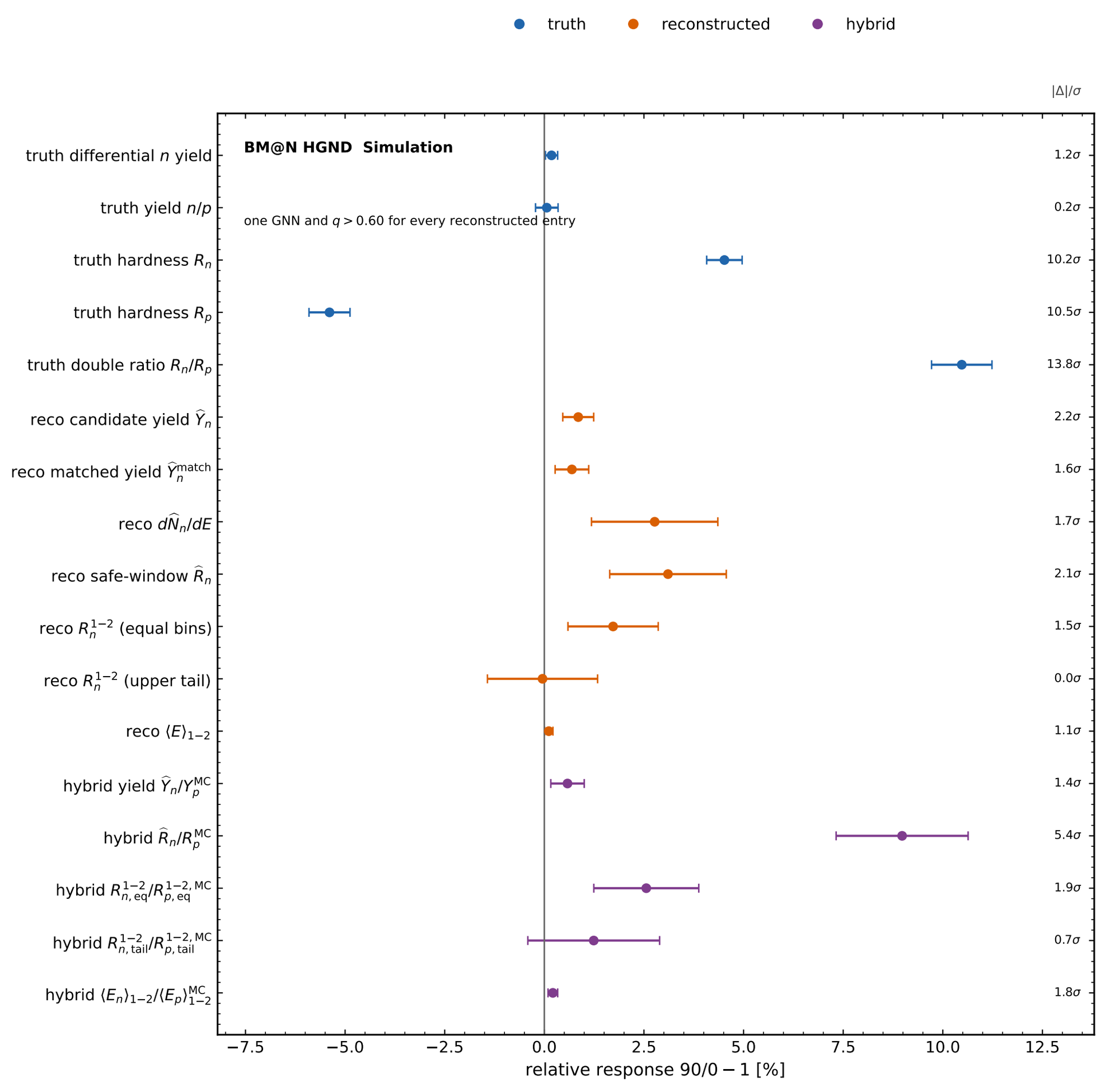
truth
reconstructed
hybrid
|Δ|/σ
BM@N HGND Simulation
one GNN and q > 0.60 for every reconstructed entry
truth differential n yield
truth yield n/p
truth hardness R_n
truth hardness R_p
truth double ratio R_n/R_p
reco candidate yield $\widehat{Y}_n$
reco matched yield $\widehat{Y}_n^{\mathrm{match}}$
reco $d\widehat{N}_n/dE$
reco safe-window $\widehat{R}_n$
reco R_n^{1-2} (equal bins)
reco R_n^{1-2} (upper tail)
reco $\langle E \rangle_{1-2}$
hybrid yield $\widehat{Y}_n/Y_p^{\mathrm{MC}}$
hybrid $\widehat{R}_n/R_p^{\mathrm{MC}}$
hybrid $R_{n,\mathrm{eq}}^{1-2}/R_{p,\mathrm{eq}}^{1-2,\mathrm{MC}}$
hybrid $R_{n,\mathrm{tail}}^{1-2}/R_{p,\mathrm{tail}}^{1-2,\mathrm{MC}}$
hybrid $\langle E_n \rangle_{1-2}/\langle E_p \rangle_{1-2}^{\mathrm{MC}}$
1.2σ
0.2σ
10.2σ
10.5σ
13.8σ
2.2σ
1.6σ
1.7σ
2.1σ
1.5σ
0.0σ
1.1σ
1.4σ
5.4σ
1.9σ
0.7σ
1.8σ
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
relative response 90/0 − 1 [%]